

Equivalence Completion for Neutrino Mixing and Oscillation Invariants

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Abstract

We complete an equivalence map between flavor–mass bookkeeping and observable weak-transition bookkeeping for neutrino propagation in the oscillation regime. Parameters commonly treated as independently physical in a flavor-basis parameterization are shown to overcount tensor load once admissible redescription and regime transport are enforced. Only mass-squared differences and rephasing-invariant mixing combinations survive as invariant witnesses. The result is eliminative, invariant under admissible redescription, and introduces no new physics.

1 Apparent Independence in Flavor–Mass Bookkeeping

Let flavor states ν_α be related to mass eigenstates ν_i by a unitary matrix U ,

$$\nu_\alpha = \sum_i U_{\alpha i} \nu_i. \quad (1)$$

In flavor–mass bookkeeping, one parameterizes U by angles and phases and treats these parameters together with masses $\{m_i\}$ as independent physical inputs.

2 Observable Transition Bookkeeping

In the oscillation regime, transition amplitudes depend on interference between mass eigenstates. For baseline L and energy E , the transition probability takes the form

$$P_{\alpha \rightarrow \beta} = \sum_{i,j} U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j} \exp\left(-i \frac{\Delta m_{ij}^2 L}{2E}\right), \quad (2)$$

where

$$\Delta m_{ij}^2 \equiv m_i^2 - m_j^2. \quad (3)$$

3 Admissible Redescription: Rephasing

Consider admissible flavor and mass rephasings,

$$\nu_\alpha \rightarrow e^{i\phi_\alpha} \nu_\alpha, \quad \nu_i \rightarrow e^{i\psi_i} \nu_i. \quad (4)$$

These induce

$$U_{\alpha i} \rightarrow e^{i\phi_\alpha} U_{\alpha i} e^{-i\psi_i}. \quad (5)$$

Under these admissible redescription, the quantities

$$U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j} \quad (6)$$

are invariant, while many individual parameterization phases are not.

4 Lemma: Invariant Witness Counting

If a tensor load contains n independent degrees of freedom, admissible redescription and regime transport must preserve n independent invariant witnesses across all admissible skins describing the same regime. A parameter that can be varied without changing any invariant witness is redundancy and has no standing as an independent tensor degree.

5 Failure of Independence

Assume that the full parameterization degrees of freedom of U (angles and phases) together with $\{m_i\}$ correspond to independent tensor degrees. Observable transition bookkeeping exposes only:

- mass-squared differences Δm_{ij}^2 (not absolute mass offsets), and
- rephasing-invariant products $U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}$.

No additional independent invariant witnesses exist. Preservation of the assumed independence would require hidden unwitnessed structure, loss of tensor load under transport, or representational privilege. Each constitutes standing failure.

6 Forced Tensor Relation

The neutrino oscillation tensor load is uniquely specified by the invariant set:

$$\boxed{\Delta m_{ij}^2 \equiv m_i^2 - m_j^2} \quad (7)$$

together with the rephasing-invariant mixing factors

$$\boxed{U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}} \quad (8)$$

appearing in

$$\boxed{P_{\alpha \rightarrow \beta} = \sum_{i,j} U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j} \exp\left(-i \frac{\Delta m_{ij}^2 L}{2E}\right)} \quad (9)$$

Any parameterization freedom in U beyond these rephasing invariants is redundancy. Absolute mass offsets are not invariant witnesses in this skin.

7 Corollaries

- Rephasing-removable phases in U have no standing as independent tensor degrees.
- Any two parameter sets producing the same Δm_{ij}^2 and the same rephasing-invariant products are tensor-identical for oscillation phenomena.

- Any framework asserting additional independently tunable observable transition degrees beyond Δm^2 and rephasing invariants violates standing unless it introduces forbidden tensor load.

Conclusion

Equivalence completion collapses apparent freedom in neutrino mixing parameterizations to an invariant witness set. The result is eliminative, representation-invariant, and irreversible under admissible redescription.